TECHNICAL VALIDATION

Email & SMS Management Plan Validation Report

Date: 2026-07-07

Reviewer scope: Full validation across the dynamic-scheduling track (story + implementation plan), the Dynamic Recipient Resolution (DRR, story 77.9) and SMS provider integration (story 76.8) gap analyses, the known-issues registers, the user stories, and the consolidated template list.

Method: Deep read of all supplied documents, cross-checked against industry best practices for scheduled-notification systems, the transactional-outbox / at-least-once dispatch pattern, daylight saving time (DST)-safe scheduling, 2026 US SMS compliance (A2P 10DLC registration and TCPA rules), and scheduled-send recipient-resolution semantics (sources listed at the end).

Deliverable form: Validation report + fixes. **Your original plan files are left untouched** — apply the fixes below into them at your discretion.

> **Verdict:** The plan is sound and above the bar for this class of feature. Approve to build, incorporating the fixes listed in Section 3.

1. Verdict

> **The plan is sound and above the bar for this class of feature. Approve to build, incorporating the fixes listed in Section 3.**

The core engineering shape is correct by industry standards and needs no redesign:

- **Materialize occurrences into a table with a stable, non-time dedupe key**, recompute `fire_at` for PENDING rows and freeze it for SENT rows — this is exactly the idempotent, at-least-once model the literature prescribes, and the choice of `offset_key`/`sequence_index` (never `fire_at`) for identity is the subtle detail most teams get wrong.
- **Atomic claim (`updateMany WHERE status=PENDING SET SENDING`) + SENDING-reaper + retry/backoff + bounded series** mirror the transactional-outbox playbook (claim, lease, poison-message handling).
- **DST-correct wall-clock via `date-fns-tz` with location-based IANA zones**, plus explicit spring-forward gap handling, matches the current best practice (store UTC as source of truth, compute from local wall-clock + IANA zone, never fixed UTC offsets).
- **Shipping around the two deferred dependencies with hard, mechanical gates** (SMS: materialize then SKIP; token recipients: SKIP with reason "requires DRR"; column/one-hop recipients ship now) is a disciplined way to deliver the engine without waiting on DRR and SMS.
- **`is_schedulable` as a first-class marked flag gated by a code-owned trigger ceiling**, and **by-id dispatch making known issue #21 immunity true by construction**, are both correct and well-argued.
- The **DRR and SMS gap analyses are genuinely strong** — decision-ready question registers with file:line evidence and correct owner tagging. Little to fix in their substance; the fixes below are mostly about *sequencing* and *one factual doc correction*.

What follows are the issues worth fixing before/while building. None invalidate the architecture. Severity: **High** = fix before that phase ships; **Medium** = fix during the phase; **Low** = tidy-up / documentation.

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2. Issue summary (at a glance)

ID	Area	Severity	One-line
S1	Scheduling	**HIGH**	No retention/purge job for the high-volume occurrences table — it grows forever.
S2	Scheduling	Medium	Exactly-once depends on single-instance topology; reaper race can double-send with no provider idempotency.
S3	Scheduling	Medium	24h catch-up skip is one-size-fits-all; can silently drop a still-valid due send.
S4	Scheduling	Medium	RECURRING "until answered" has no defined instance-discovery query and depends on a deferred-dependency resolver.
S5	Scheduling	Low	Confirm DST **ambiguous** (fall-back) case is unit-tested, not just the gap.
S6	Scheduling	Medium	Free-form event timezone silently defaults on invalid input, causing wrong-hour sends. Fail closed instead.
S7	Scheduling	Low	`recipients_snapshot` stores personally identifiable information (PII) with no retention tie-in (addressed by finding S1).
D1	DRR	**HIGH**	Resolve-timing conflict (DRR gap item 13) — resolve it as a **per-rule toggle**, not a global decision.
D2	DRR	Medium	`Exhibitor.company_id @unique` schema drift is a latent bug beyond DRR — fix now.
D3	DRR	Medium	Zero-recipient fallback must not silently skip **transactional** sends.
M1	SMS	**HIGH**	SMS doc-vs-code contradiction (SMS gap item 15) — correct the known-issues text before the client sees it.
M2	SMS	Medium	Quiet-hours/consent model is under-specified for 2026 rules (state-aware hours, 10DLC gate).
M3	SMS	Medium	`NotificationLog` cannot record an SMS destination/channel — must land before any SMS dispatch.
M4	SMS	Low	DRR and SMS circular dependency — break it by sequencing email DRR first.
X1	Cross-doc	Low	Drop `CART_CONVERTED` from the enum until a distinct state exists.
X2	Cross-doc	Low	Add retention / double-send / bad-timezone cases to the verification list.

3. Findings and fixes

Scheduling track

HIGH S1 — NO RETENTION/PURGE FOR `NOTIFICATION_SCHEDULE_OCCURRENCES`

CURRENT STATE

FORWARD GROWTH IS BOUNDED WELL: `SCHEDULE_MATERIALIZE_HORIZON_DAYS` (45), `SCHEDULE_RECURRING_HORIZON_DAYS` (14), ONE-OCCURRENCE-AT-A-TIME PER RECURRING INSTANCE. THAT CAPS HOW FAR *AHEAD* ROWS ARE CREATED.

GAP IDENTIFIED

NOTHING PURGES ROWS ONCE THEY REACH A TERMINAL STATE (`SENT/SKIPPED/CANCELLED/FAILED`). THIS IS A `BIGINT` PRIMARY KEY, EXPLICITLY "HIGH-VOLUME" TABLE. THE OUTBOX LITERATURE IS UNANIMOUS THAT THE PROCESSED-ROW TABLE NEEDS A SCHEDULED RETENTION JOB OR IT BECOMES AN OPERATIONAL PROBLEM (BLOAT, SLOW `(STATUS, FIRE_AT)` SCANS, BACKUP WEIGHT). IT IS THE SINGLE MOST COMMON OMISSION IN EXACTLY THIS DESIGN.

RECOMMENDED FIX

ADD A SECOND, LOW-FREQUENCY CRON (REUSE THE SAME REGISTRAR PATTERN) THAT BATCH-DELETES OR ARCHIVES TERMINAL-STATUS OCCURRENCES OLDER THAN A NEW PPL_SETTINGS `SCHEDULE_OCCURRENCE_RETENTION_DAYS` (SUGGEST 90, CLAMPED). USE BATCHED DELETES (E.G. `DELETE ... WHERE ID IN (SELECT ID ... LIMIT 5000)` IN A LOOP), NEVER A SINGLE BLOCKING DELETE, PER THE RETENTION BEST PRACTICE. IF ANY OCCURRENCE AUDIT MUST BE RETAINED LONGER, ARCHIVE TO `NOTIFICATIONLOG` (WHICH IS ALREADY THE PERMANENT AUDIT) BEFORE DELETING THE OCCURRENCE. THIS ALSO RESOLVES S7 (PII IN `RECIPIENTS_SNAPSHOT` IS THEN TIME-BOUNDED).

MEDIUM S2 — EXACTLY-ONCE LEANS ON TOPOLOGY; REAPER RACE CAN DOUBLE-SEND

CURRENT STATE

CORRECT AT-LEAST-ONCE MECHANICS: ATOMIC PER-ROW CLAIM, SENDING-REAPER RESETS CRASHED CLAIMS, `DEDUPE_KEY` UNIQUE PREVENTS DUPLICATE *MATERIALIZATION*, AND THE WORKER RUNS SINGLE-INSTANCE (`INSTANCES:1, MAXSURGE:0`).

GAPS IDENTIFIED

- THE PLAN STATES THE TOPOLOGY *IS* THE CROSS-PROCESS GUARD. THAT IS TRUE ONLY WHILE THE DEPLOY GENUINELY NEVER OVERLAPS — `MAXSURGE:0` REDUCES BUT DOES NOT GUARANTEE ZERO OVERLAP ON EVERY PLATFORM, AND ANY FUTURE SCALE-OUT SILENTLY BREAKS THE ASSUMPTION. THE ATOMIC `UPDATEMANY` CLAIM ACTUALLY *DOES* HOLD UNDER CONCURRENCY, SO THE REAL EXPOSURE IS NARROWER THAN "NEEDS SINGLE INSTANCE" — WORTH STATING PRECISELY.
- THE GENUINE RESIDUAL RISK IS THE REAPER-VS-SLOW-SEND DOUBLE-SEND: THE REAPER FLIPS A STILL-IN-FLIGHT `SENDING` ROW BACK TO `PENDING` AFTER `SCHEDULE_SENDING_STALE_MINUTES` (15), A SECOND TICK RE-DISPATCHES, AND SENDGRID HAS NO IDEMPOTENCY KEY BY DEFAULT, SO THE RECIPIENT CAN GET TWO EMAILS. THE `DEDUPE_KEY` DOES NOT PREVENT THIS — IT DEDUPES MATERIALIZATION, NOT DISPATCH.

RECOMMENDED FIX

- ADOPT `SELECT ... FOR UPDATE SKIP LOCKED` FOR THE CLAIM (POSTGRES-NATIVE, THE STANDARD OUTBOX CLAIM) SO CORRECTNESS STOPS DEPENDING ON DEPLOYMENT TOPOLOGY. IT COMPOSES WITH THE EXISTING `UPDATEMANY` STATUS GUARD.
- SET `SCHEDULE_SENDING_STALE_MINUTES` COMFORTABLY ABOVE THE WORST-CASE SENDGRID CALL + NETWORK TIMEOUT (15 MIN IS ALREADY GENEROUS — KEEP IT, BUT DOCUMENT *WHY* THE FLOOR EXISTS).
- STATE EXPLICITLY IN THE PLAN THAT THE SYSTEM IS AT-LEAST-ONCE, NOT EXACTLY-ONCE, AND THAT THE REAPER RACE IS THE ACCEPTED TAIL RISK. IF A STRONGER GUARANTEE IS WANTED LATER, PASS A SENDGRID IDEMPOTENCY MECHANISM OR A `NOTIFICATIONLOG` PRE-INSERT KEYED ON `DEDUPE_KEY` AS THE SEND-SIDE DEDUPE. FOR A TRANSACTIONAL-EMAIL VOLUME THIS IS USUALLY ACCEPTABLE TO LEAVE AS AT-LEAST-ONCE — JUST MAKE IT A STATED DECISION, NOT AN IMPLIED GUARANTEE.

MEDIUM S3 — ONE-SIZE CATCH-UP SKIP CAN DROP A LEGITIMATELY-DUE SEND

CURRENT STATE

AFTER DOWNTIME, OCCURRENCES OLDER THAN `SCHEDULE_DISPATCH_MAX_CATCHUP_MINUTES` (DEFAULT 1440 = 24H) ARE SKIPPED `"MISSED SEND WINDOW"`.

GAP IDENTIFIED

24H IS CORRECT FOR A PROXIMITY REMINDER (A "7 DAYS BEFORE EVENT" EMAIL SENT 30H LATE IS NOISE). IT IS *NOT* OBVIOUSLY CORRECT FOR EVERY KIND — E.G. A PAYMENT-DUE OR CONTRACT-REMINDER OCCURRENCE THAT CAME DUE DURING A 26H OUTAGE ARGUABLY STILL SHOULD SEND. A SINGLE GLOBAL WINDOW BAKES ONE POLICY INTO ALL KINDS.

RECOMMENDED FIX

MAKE THE CATCH-UP POLICY SELECTABLE — EITHER A PER-KIND DEFAULT (ANCHOR_RELATIVE PROXIMITY: SKIP; FOLLOW_UP/PAYMENT: SEND ANYWAY) OR A PER-RULE `CATCHUP_POLICY` OF EITHER SKIP OR SEND, WITH THE 24H DEFAULT AS THE SKIP THRESHOLD. AT MINIMUM, EMIT AN ALERT/LOG LINE FOR EVERY OCCURRENCE SKIPPED BY THE CATCH-UP SWEEP SO A SILENT MASS-SKIP AFTER AN OUTAGE IS VISIBLE, NOT INVISIBLE.

MEDIUM S4 — RECURRING "UNTIL ANSWERED" INSTANCE-DISCOVERY IS UNSPECIFIED

CURRENT STATE

ANCHOR_RELATIVE SCANS ANCHOR ROWS IN A LOOK-AHEAD WINDOW. RECURRING MATERIALIZES PER-SCHEDULE (CALENDAR) OR PER-INSTANCE ("UNTIL ANSWERED") ONE OCCURRENCE AT A TIME, BOUND BY `ANCHOR_INSTANCE_REF='ORDER_PRODUCT:ID'`.

GAP IDENTIFIED

FOR THE PER-INSTANCE "UNTIL ANSWERED" CASE, THE PLAN NEVER DEFINES THE QUERY THAT *DISCOVERS WHICH INSTANCES CURRENTLY NEED A SERIES* (WHICH `ORDER_PRODUCT` ROWS HAVE UNANSWERED QUESTIONS). AND ITS STOP-CONDITION, `QUESTION_ANSWERED`, IS ITSELF A DEFERRED DEPENDENCY (ANSWER TABLE NOT MODELLED). SO THE ONE CLIENT RECURRING-WITH-STOP USE CASE HAS NEITHER A START QUERY NOR A LIVE STOP RESOLVER.

RECOMMENDED FIX

EITHER (A) SPECIFY THE INSTANCE-DISCOVERY QUERY AND MARK THE WHOLE "UNANSWERED PRODUCT QUESTIONS" RECURRING TEMPLATE DEFERRED ALONGSIDE ITS DEFERRED-DEPENDENCY ANCHOR/RESOLVER (CLEANEST — IT MATCHES HOW THE SHOW/WORKSHOP ANCHORS ARE ALREADY DEFERRED), OR (B) IF IT MUST SHIP, DEFINE THE ENUMERATING QUERY AND REQUIRE AN `END_WINDOW_AT` BOUND UNTIL `QUESTION_ANSWERED` LANDS (THE PLAN'S HARD RULE ALREADY IMPLIES THIS — MAKE IT EXPLICIT FOR THIS TEMPLATE). RECOMMEND (A) FOR THIS BUILD; IT KEEPS PHASE 4 HONEST.

LOW S5 — CONFIRM DST AMBIGUOUS-TIME TEST COVERAGE

FINDING

THE PLAN HANDLES THE SPRING-FORWARD GAP (NORMALIZE FORWARD) AND PICKS THE EARLIER INSTANT FOR THE FALL-BACK AMBIGUOUS HOUR. BOTH CHOICES ARE FINE. THE VERIFICATION LIST ONLY NAMES A GAP TEST. ADD AN EXPLICIT UNIT TEST FOR THE AMBIGUOUS (FALL-BACK) CASE TOO, SO THE DETERMINISTIC "EARLIER OCCURRENCE" CHOICE IS PINNED.

MEDIUM S6 — INVALID EVENT TIMEZONE SILENTLY DEFAULTS, CAUSING WRONG-HOUR SEND

CURRENT STATE

`TIMEZONE='EVENT'` RESOLVES FROM `ANCHOR.TIMEZONE`; `SHOWS.TIMEZONE` IS FREE-FORM `VARCHAR(50)`; ON INVALID IANA THE PLAN DEFAULTS (TO `SCHEDULE_DEFAULT_TIMEZONE`, AMERICA/NEW_YORK) WITH A WARNING.

GAP IDENTIFIED

SILENTLY DEFAULTING A MALFORMED EVENT TIMEZONE MEANS A SEND FIRES AT THE WRONG LOCAL HOUR WITH ONLY A LOG WARNING — EXACTLY THE KIND OF "LOOKS FINE, WRONG BY AN HOUR" BUG DST HANDLING IS MEANT TO PREVENT. THE CLIENT'S NOTE-47 TIMEZONE-FIDELITY REQUIREMENT IS THE WHOLE POINT.

RECOMMENDED FIX

VALIDATE/NORMALIZE `SHOWS.TIMEZONE` TO A CANONICAL IANA ZONE AT WRITE/INGEST TIME AND REJECT OR FLAG INVALID VALUES THERE. AT SEND TIME, IF EVENT RESOLUTION YIELDS AN INVALID/AGAIN-NULL ZONE, PREFER FAIL-CLOSED (OCCURRENCE SKIPPED + ALERT) OVER SILENTLY SENDING AT A GUESSED ZONE — A NOT-SENT, SURFACED REMINDER IS SAFER THAN A WRONG-TIME ONE. KEEP THE DEFAULT ONLY FOR THE EXPLICITLY-CHOSEN (NON-EVENT) PATH.

LOW S7 — PERSONALLY IDENTIFIABLE INFORMATION (PII) IN `RECIPIENTS_SNAPSHOT`

FINDING

THE PLAN FLAGS THIS AS OPEN IN SECTION 9 OF THE IMPLEMENTATION PLAN. IT IS ACCEPTABLE GIVEN IT ENABLES THE NO-DRR SHIP PATH, AND IT BECOMES PROPERLY BOUNDED ONCE S1 (RETENTION) LANDS. RECOMMEND EXPLICITLY TYING `RECIPIENTS_SNAPSHOT` LIFETIME TO THE OCCURRENCE RETENTION WINDOW AND NOTING THE DATA-MINIMIZATION RATIONALE. NO SEPARATE WORK BEYOND S1.

DRR (story 77.9)

HIGH D1 — RESOLVE THE TIMING CONFLICT (DRR GAP ITEM 13) AS A PER-RULE TOGGLE

CONFLICT

STORY 77.9 SAYS "RESOLVE RECIPIENTS AT SEND TIME USING THE MOST CURRENT DATA." THE SCHEDULER RESOLVES AT MATERIALIZE/CAPTURE TIME AND REPLAYS A SNAPSHOT VERBATIM. FOR A MULTI-DAY SCHEDULED SEND, A SALESPERSON REASSIGNED OR A CONTACT REMOVED AFTER MATERIALIZE STILL GETS THE EMAIL.

BEST-PRACTICE RESOLUTION

THIS IS A SOLVED PRODUCT PROBLEM, NOT AN ARCHITECTURE CLASH. MATURE SENDERS (E.G. KLAVIYO) EXPOSE IT AS A PER-CAMPAIGN SWITCH — "DETERMINE RECIPIENTS AT SEND TIME" ON/OFF — AND, CRUCIALLY, DOCUMENT THAT RECIPIENT-LOCAL TIMEZONE SENDING IS INCOMPATIBLE WITH SEND-TIME RESOLUTION AND FORCES THE SNAPSHOT APPROACH. THAT IS PRECISELY THE SCHEDULER'S SITUATION.

RECOMMENDED FIX

MAKE IT AN EXPLICIT, PER-RULE CHOICE RATHER THAN A SINGLE GLOBAL DECISION:

- DEFAULT = SNAPSHOT AT MATERIALIZE/CAPTURE (WHAT THE SCHEDULER ALREADY DOES; MATCHES TIMEZONE-ACCURATE SENDS).
- OPTIONAL `RESOLVE_AT_SEND: BOOLEAN` ON A RULE FOR CASES THAT NEED FRESHNESS; WHEN SET, THE OCCURRENCE STORES A *REFERENCE* (ANCHOR ID + TOKEN SPEC) INSTEAD OF A RESOLVED SNAPSHOT, AND DRR RESOLVES AT DISPATCH. DOCUMENT THAT THIS PATH IS UNAVAILABLE UNTIL DRR SHIPS, AND IS MUTUALLY EXCLUSIVE WITH THE PRE-MATERIALIZED RECIPIENT SNAPSHOT.

THIS CONVERTS DRR GAP ITEM 13 FROM A BLOCKING CONFLICT INTO A DOCUMENTED PRODUCT OPTION, AND LETS THE BUSINESS ANALYST ANSWER THE CHECKLIST QUESTION WITH "BOTH, SELECTABLE" INSTEAD OF CHOOSING ONE BEHAVIOR FOR EVERYONE. IT ALSO MEANS THE SCHEDULER NEEDS NO REDESIGN — IT ALREADY IMPLEMENTS THE DEFAULT.

MEDIUM D2 — `EXHIBITOR.COMPANY_ID @UNIQUE` DRIFT IS A LATENT BUG BEYOND DRR

FINDING

THE GAP ANALYSIS CORRECTLY SURFACES THAT `@UNIQUE` IN BOTH SCHEMAS CONTRADICTS THE REAL NON-UNIQUE DB INDEX, AND THAT MULTI-MEMBER IS REAL. THIS IS NOT JUST A DRR BLOCKER — ANY PRISMA QUERY THAT ASSUMES ONE EXHIBITOR PER COMPANY CAN SILENTLY REGRESS (FETCH-ONE WHERE THE DB HAS MANY). RECOMMEND FORMALLY DROPPING `@UNIQUE` FROM BOTH SCHEMA FILES AS A STANDALONE DATA-INTEGRITY FIX, DECOUPLED FROM DRR TIMING, SO IT CAN'T BITE AN UNRELATED QUERY FIRST.

MEDIUM D3 — ZERO-RECIPIENT FALLBACK MUST NOT SILENTLY SKIP TRANSACTIONAL SENDS

FINDING

DRR GAP ITEM 6 LEAVES THE FALLBACK "SUBJECT TO RESEARCH AND DEVELOPMENT" (SKIP / DEFAULT / ABORT). THE SCHEDULER'S PRECEDENT IS SKIP-AND-LOG, WHICH IS RIGHT FOR REMINDERS. BUT APPLYING SKIP-SILENTLY TO A TRANSACTIONAL TRIGGER (ORDER CONFIRMATION, REFUND) MEANS A CUSTOMER-CRITICAL EMAIL VANISHES WITH ONLY A LOG LINE. RECOMMEND THE BUSINESS ANALYST ANSWER LAND AS: SKIP-AND-LOG FOR MARKETING/REMINDER TRIGGERS, ABORT-AND-SURFACE (ALERT) FOR TRANSACTIONAL TRIGGERS, NEVER SEND TO ZERO RECIPIENTS. TIE THE "SURFACE" TO THE SAME ALERTING ADDED IN S3.

SMS (story 76.8)

HIGH M1 — CORRECT THE SMS DOC-VS-CODE CONTRADICTION (SMS GAP ITEM 15) BEFORE THE CLIENT SEES IT

FINDING

`EMAIL_SMS_KNOWN_ISSUES.MD` (KNOWN ISSUES #2 AND #12) STATES SMS STORAGE/EDIT IS "ALREADY BUILT, ZERO SCHEMA CHANGE," BUT THE CODE HARD-BLOCKS SMS CREATE (`SUPPORTED_TEMPLATE_CHANNELS=['EMAIL']`, SERVICE THROWS), SEEDS ZERO SMS ROWS, AND LEFT THE SMS `CHANNEL_CONFIG` SHAPE READ-ONLY-BUT-UNDEFINED. THE GAP ANALYSIS ALREADY CAUGHT THIS. FIX THE KNOWN-ISSUES TEXT SO IT READS "SMS CREATE IS GATED; STORAGE SHAPE UNDEFINED" — OTHERWISE THE REGISTER OVERSTATES READINESS AND A CLIENT OR PROJECT MANAGER COULD SCOPE SMS AS NEARLY-DONE WHEN IT IS NOT. THIS IS A FIVE-MINUTE DOC EDIT WITH OUTSIZED CREDIBILITY IMPACT.

MEDIUM M2 — 2026 SMS COMPLIANCE MODEL IS UNDER-SPECIFIED

FINDING

SMS GAP ITEM 3 CITES QUIET HOURS "8AM–9PM" AND LEANS ON THE PROVIDER FOR STOP. 2026 REALITY IS STRICTER AND WORTH BAKING INTO THE ANSWER THE BUSINESS ANALYST AND CLIENT GIVE:

- QUIET HOURS ARE NOW STATE-AWARE — FL/OK/WA EFFECTIVELY 8AM–8PM; TEXAS (FROM SEP 2025) 9AM–9PM MON–SAT AND NOON–9PM SUN. A FLAT 8–9 WINDOW IS NOT SUFFICIENT FOR A NATIONAL AUDIENCE.
- OPT-OUT MUST BE HONORED VIA "ANY REASONABLE METHOD," NOT ONLY THE STOP KEYWORD THE PROVIDER AUTO-HANDLES — SO A SUPPRESSION STORE THE PLATFORM CONTROLS IS NEEDED REGARDLESS OF PROVIDER.
- CONSENT RECORDS SHOULD BE RETAINED FOR AT LEAST 5 YEARS.
- HARD GATE: SINCE FEB 2025 CARRIERS *BLOCK* (NOT THROTTLE) 100% OF UNREGISTERED 10DLC TRAFFIC. ADD AN EXPLICIT PRE-LAUNCH CHECKLIST ITEM: *NO SMS IN PRODUCTION UNTIL THE 10DLC BRAND + CAMPAIGN ARE REGISTERED AND THE SENDING NUMBER/MESSAGING SERVICE IS PROVISIONED.*

NONE OF THIS CHANGES THE "SMS DEFERRED" DECISION — IT SHARPENS SMS GAP ITEMS 3 AND 8 SO THAT WHEN SMS IS PULLED FORWARD, THE COMPLIANCE SURFACE IS SCOPED CORRECTLY RATHER THAN DISCOVERED LATE.

MEDIUM M3 — `NOTIFICATIONLOG` MUST GAIN CHANNEL + DESTINATION BEFORE ANY SMS DISPATCH

FINDING

SMS GAP ITEM 5 IS CORRECT: THE SINGLE `EMAIL STRING?` COLUMN CANNOT RECORD AN SMS DESTINATION, AND THERE IS NO CHANNEL DISCRIMINATOR. RECOMMENDATION ON THE OPEN SHAPE QUESTION: EXTEND `NOTIFICATIONLOG` WITH `CHANNEL` AND A GENERALIZED `RECIPIENT` COLUMN (ADMIN-OWNED MIGRATION, PROPAGATED VIA `DB PUSH`) RATHER THAN A SEPARATE SMS LOG TABLE — IT KEEPS ONE AUDIT SURFACE, ONE QUERY PATH, AND MATCHES THE "GENERALIZE EMAIL TO RECIPIENT" DIRECTION. THIS IS A PREREQUISITE FOR SMS DISPATCH AND SHOULD BE SEQUENCED IMMEDIATELY AFTER THE SCOPE DECISION (SMS GAP ITEM 2), NOT ALONGSIDE SEND CODE.

LOW M4 — BREAK THE DRR AND SMS CIRCULAR DEPENDENCY BY SEQUENCING

FINDING

SMS GAP ITEM 7 AND DRR GAP ITEM 9 NOTE THAT STORY 76.8 DEPENDS ON DRR FOR PHONE RESOLUTION WHILE DRR IS EMAIL-ONLY AND BOTH ARE DEFERRED — A CIRCULAR BLOCK. RECOMMEND STATING THE SEQUENCE EXPLICITLY: (1) SHIP EMAIL DRR GENERALIZING THE SCHEDULER'S RESTRICTED RESOLVER; (2) SMS REUSES THAT RESOLVER, EXTENDED TO A PHONE FIELD, RATHER THAN STANDING UP A PARALLEL MECHANISM. THAT ORDERING (ALREADY IMPLIED BY THE DRR GAP ANALYSIS'S "EXTEND THE ALLOW-LIST, DON'T FORK") REMOVES THE CIRCULARITY ON PAPER.

Cross-document / process

LOW X1 — DROP `CART_CONVERTED` UNTIL A DISTINCT STATE EXISTS

FINDING

THE PLAN ITSELF NOTES `CONTRACT_SIGNED` AND `CART_CONVERTED` ARE ALIASES ON TODAY'S SCHEMA (CART CONVERTS TO ORDER ON SIGNATURE). SHIPPING BOTH AS ENUM VALUES IMPLIES A CAPABILITY THAT DOESN'T EXIST AND INVITES A FUTURE AUTHOR TO USE `CART_CONVERTED` EXPECTING DIFFERENT SEMANTICS. RECOMMEND THE PLAN'S OWN ALTERNATIVE: DROP `CART_CONVERTED` FROM THE ENUM UNTIL A GENUINELY DISTINCT STATE IS MODELLED, KEEPING THE STOP-CONDITION SET HONEST.

LOW X2 — EXTEND THE VERIFICATION LIST

FINDING

THE VERIFICATION SECTION IS ALREADY THOROUGH. ADD THREE CASES MATCHING THE FIXES ABOVE:

- RETENTION — TERMINAL-STATUS OCCURRENCES OLDER THAN THE RETENTION WINDOW ARE PURGED IN BATCHES; ACTIVE/PENDING UNTOUCHED (S1).
- REAPER DOUBLE-SEND — A STUCK `SENDING` OCCURRENCE PAST THE STALE WINDOW IS RETRIED AND, GIVEN AT-LEAST-ONCE, THE ACCEPTED BEHAVIOR IS DOCUMENTED AND IDEMPOTENCY-AT-PROVIDER (IF ADOPTED) PREVENTS A DUPLICATE (S2).
- BAD TIMEZONE — AN INVALID `SHOWS.TIMEZONE` IS REJECTED AT INGEST / FAILS CLOSED AT SEND (SKIPPED + ALERT), NEVER SILENTLY DEFAULTED (S6).

4. What is explicitly right (keep as-is)

So the fixes above are read as polish, not alarm — these decisions are correct and should not be second-guessed:

- Dedicated tables over the JSON columns; JSON columns repurposed as advisory author hints only.
- Stable `offset_key/sequence_index` identity; PENDING-recompute / SENT-immutable split (Acceptance Criteria 3 and 4).
- Multi-offset look-ahead window keyed to the union of offsets; RECURRING bounded roll-forward; FOLLOW_UP fixed `series_anchor_at` so late sends don't drift the series.
- DST-correct wall-clock with `date-fns-tz` and IANA zones; rejecting `timezone='EVENT'` for RECURRING and for anchors with no timezone column.
- Restricted `recipient_source` / `replacements_map` resolver (bare column or one documented relation hop + fixed named transforms), validated at config **and** materialize time — **no expression domain-specific language and no eval**. This is the right security posture.
- `is_schedulable` marked (not inferred), gated by the `supports_scheduling` trigger ceiling; all 18 seeded rows correctly `false`.
- By-id `notificationTemplateId` dispatch making known issue #21 immunity true by construction, shipped together with the slug-path fix.
- The materialize-then-SKIP gates for SMS and token-recipients — clean, zero-further-schema deferral.
- The DRR/SMS gap analyses' discipline: evidence-backed, owner-tagged, decision-ready.

5. Recommended sequence for the fixes

1. **Before Phase 3 ships:** S1 (retention job), S6 (timezone fail-closed), M1 (doc correction —

do immediately, it's a text edit).

2. **During Phase 3/4:** S2 (SKIP LOCKED + at-least-once statement), S3 (catch-up policy + alerting), S4 (defer or specify "until answered"), X1 (drop enum value), X2 (tests).

3. **Decoupled, do soon:** D2 (drop `@unique` drift) — independent of DRR.

4. **Feed into the business analyst and client sessions (no build yet):** D1 (per-rule resolve toggle), D3 (transactional zero-recipient policy), M2 (2026 compliance scope), M3 (NotificationLog shape), M4 (DRR-then-SMS sequence).

6. Open questions for you (only if you want them resolved before I revise anything further)

None are blockers for this report; flagging where your intent would change a recommendation:

Exactly-once vs at-least-once (S2) — is at-least-once with the reaper tail-risk acceptable for these emails, or do you want a provider-side idempotency mechanism designed in now?

"Unanswered product questions" RECURRING (S4) — defer it with the show/workshop anchors, or must it ship this build?

Resolve-timing (D1) — happy to adopt the per-rule toggle as the recommendation to the business analyst, or do you want to hold the pure snapshot model and simply document the staleness?

Sources

- Idempotency & Deduplication — System Design Sandbox (https://www.systemdesignsandbox.com/learn/idempotency-deduplication)
- Idempotent requests in notification infrastructure — Fyno (https://www.fyno.io/blog/idempotent-requests-in-notification-infrastructure-cm4s7axck002x9jffvml6fx1y)

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- Implementing the Outbox Pattern — Milan Jovanović (https://milanjovanovic.tech/blog/implementing-the-outbox-pattern)
- Handling Daylight Saving Time in Applications: A Developer's Complete Guide (https://www.datetimeapp.com/learn/handling-daylight-saving-time)
- Cron Timezone and DST Cheat Sheet — ClientOps (https://clientops.dev/guides/cron-timezone-dst-cheat-sheet/)
- 2026 Guide to TCPA Compliance for SMS in the US — Infobip (https://www.infobip.com/blog/tcpa-compliance-sms)
- A2P 10DLC guide: 2026 US compliance and regulations — Infobip (https://www.infobip.com/blog/what-is-a2p-10dlc)
- SMS compliance in 2026: What to know before you send — Telnyx (https://telnyx.com/resources/sms-compliance)
- Understanding campaign schedule and send options — Klaviyo Help Center (https://help.klaviyo.com/hc/en-us/articles/360050216012)